SEMANTIC AND SIMILARITY-BASED ANSWER EVALUATION SYSTEM USING NLP

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**SEMANTIC AND SIMILARITY – BASED ANSWER EVALUATION SYSTEM USING NLP"** has been carried out by the students **KISHORE M, NACHIAPPAN PL, RAKSHANA G, RASEEN RAJA R.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Evaluation of descriptive answers in examinations is usually done manually by teachers, which requires significant time and effort. Manual evaluation may also lead to inconsistency and bias in grading. To overcome these issues, this project proposes a Semantic and Similarity-Based Answer Evaluation System using Natural Language Processing (NLP). The system automatically evaluates student answers by comparing them with predefined model answers. It uses text preprocessing techniques such as tokenization, stop-word removal, and lemmatization to clean the input text. Feature extraction methods like BERT embeddings are used to capture the semantic meaning of the text. The similarity between student answers and model answers is calculated using cosine similarity. Based on the similarity score, the system assigns marks automatically. This approach ensures fair, accurate, and consistent evaluation of subjective answers. The system also reduces manual workload and improves the speed of the evaluation process. It can be effectively used in online examinations and modern digital education systems.

Keywords: Natural Language Processing, Semantic Similarity, Automated Answer Evaluation, Text Preprocessing, BERT, Cosine Similarity.

ACKNOWLEDGEMENT

We thank our **Dr.N.VASUDEVAN,** Principal, for his valuable suggestions and support during the course of my research work.

We thank our **Mr.R.RAJAVARMAN,** Head of the Department, Assistant Professor (Sr. Grade), Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Mr.D.P.DEVAN,** Assistant Professor, Computer Science and Engineering for his keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs. M. MADHUMATHI,** Assistant Professor, Computer Science and Engineering, for her valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Department of Computer Science and Engineering, k Ramakrishnan College of Technology, Samayapuram, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

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LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	-	Artificial intelligence
API	-	Application programming interface
BERT	-	Bidirectional encoder representations from transformers
CSS	-	Cascading style sheets
DB	-	Database
HTML	-	Hypertext markup language
IDE	-	Integrated development environment
LMS	-	Learning management system
ML	-	Machine learning
NLP	-	Natural language processing
SQL	-	Structured query language
UI	-	User interface

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

In the traditional education system, evaluation of descriptive answers is usually performed manually by teachers or examiners. This process requires a large amount of time and effort, especially when there are many students. Manual evaluation may also lead to inconsistencies because different evaluators may assign different marks for the same answer based on their understanding or judgment. With the rapid development of Artificial Intelligence and Natural Language Processing (NLP), it has become possible for computers to understand and analyze human language more effectively. NLP techniques help machines process textual information and extract meaningful insights from written content. In recent years, automated evaluation systems have gained attention in the field of digital education and online learning platforms. These systems aim to assist teachers by reducing their workload and improving the accuracy of the evaluation process. Semantic similarity techniques allow systems to compare the meaning of two sentences rather than relying only on exact word matching. This technology can be used to compare student answers with predefined model answers and determine how closely they match in meaning. By integrating NLP techniques with machine learning models, an automated system can be developed to evaluate descriptive answers efficiently. Such systems are becoming important in modern education environments where digital assessments and online examinations are increasingly common.

1.2 OBJECTIVE

The main objective of this project is to develop an automated system that evaluates descriptive answers using Natural Language Processing techniques. The system aims to compare student answers with model answers and assign marks based on semantic similarity. One of the primary goals is to reduce the time and effort required for manual answer evaluation. Another objective is to ensure fairness and consistency in grading by minimizing human bias and subjectivity. The system also aims to analyze the meaning of the text rather than relying only on keyword matching. By using advanced techniques such as text preprocessing, feature extraction, and similarity measurement, the system can evaluate answers more intelligently. The project also focuses on improving the efficiency of the evaluation process in educational institutions. It aims to provide teachers with a simple and user-friendly interface for uploading and evaluating answers. The system is also designed to generate evaluation results quickly and accurately. Another objective is to support modern digital education platforms where automated assessments are required. Overall, the project aims to create a reliable and intelligent evaluation system that benefits both teachers and students.

1.3 OVERVIEW

The Semantic and Similarity-Based Answer Evaluation System is designed to automatically assess descriptive answers provided by students. The system works by comparing the student's answer with a predefined model answer using Natural Language Processing techniques. First, the input text is processed through a text preprocessing stage where unnecessary elements such as punctuation and special characters are removed. The text is then converted into a standardized format using techniques like tokenization and lemmatization. After preprocessing, the system extracts meaningful features

from the text using advanced language models such as BERT embeddings. These embeddings help represent the semantic meaning of the sentences in numerical form. Once the features are extracted, a similarity calculation method such as cosine similarity is used to measure how closely the student answer matches the model answer. Based on the similarity score, the system assigns marks automatically. The system can also categorize answers into correct, partially correct, or incorrect categories. A user-friendly interface allows teachers to upload answers and view results easily. The system helps improve the speed and accuracy of the evaluation process while maintaining fairness in grading.

1.4 PROBLEM STATEMENT

Evaluating descriptive answers manually is a challenging and time-consuming task for teachers, especially when dealing with a large number of students. The traditional evaluation process often requires careful reading and interpretation of each answer, which can lead to delays in result publication. In addition, manual evaluation may introduce inconsistencies because different evaluators may have different perspectives when grading answers. This can lead to unfair or biased grading in some cases. Another problem is that traditional evaluation methods mainly rely on keyword matching or exact phrases, which may not accurately reflect the meaning of the student's answer. Students may express correct ideas using different words or sentence structures, but these answers may not receive appropriate marks. With the growth of online education and digital examinations, the need for automated evaluation systems has increased significantly. Educational institutions require efficient tools to handle large-scale assessments while maintaining accuracy and fairness. Therefore, there is a need for a system that can automatically evaluate descriptive answers based on their semantic meaning rather than simple word

matching. This project addresses these challenges by developing an intelligent answer evaluation system using NLP techniques.

1.5 IMPLICATION

The implementation of an automated answer evaluation system has several important implications for the education sector. One major implication is the significant reduction in the time required to evaluate descriptive answers. Teachers can save valuable time and focus more on teaching and student development instead of spending long hours grading answer sheets. Another implication is the improvement in fairness and consistency in the grading process. Since the evaluation is performed by an automated system using predefined algorithms, the chances of human bias are minimized. The system also helps ensure that students are evaluated based on the meaning of their answers rather than exact wording. This encourages students to express their ideas in their own way while still receiving appropriate marks. Automated evaluation systems also support large-scale online examinations and digital learning platforms. Educational institutions can handle a large number of student submissions efficiently. In addition, the system can be further enhanced with features such as feedback generation and grammar analysis. Overall, the project contributes to the advancement of intelligent educational technologies and supports modern digital assessment methods.

CHAPTER 2

LITERATURE SURVEY

2.1 DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING

Ian Goodfellow, Yoshua, et al. (2020) discusses the role of deep learning techniques in Natural Language Processing applications. The authors explain how deep neural networks can process and understand large volumes of textual data. Deep learning models can capture complex relationships between words and sentences. These models are widely used in tasks such as translation, text classification, and semantic similarity analysis. The research highlights the advantages of deep learning in improving NLP performance. The paper also discusses various neural network architectures used in text processing. Deep learning models provide high accuracy in semantic analysis tasks. These techniques can be applied to automated answer evaluation systems. The study demonstrates the importance of advanced machine learning methods in modern NLP applications. It contributed significantly to the development of intelligent text analysis systems.

Furthermore, the study highlights that deep learning models have the ability to automatically learn features from large amounts of unstructured textual data without manual intervention. It suggests that training these models with large datasets improves their generalization ability. The paper also points out the importance of computational power and GPUs in training deep learning models efficiently. Additionally, it discusses the role of pre-trained models in reducing training time and improving accuracy. It also highlights challenges such as overfitting and the need for large labeled datasets. The research concludes that deep learning is a key technology for future NLP advancements. Overall, this work strengthens the development of intelligent automated evaluation systems.

2.2 AUTOMATED GRADING OF STUDENT ANSWERS USING NLP

Rahul Gupta and Ankit Sharma (2019), presents an automated grading system designed to evaluate student answers using Natural Language Processing techniques. The system processes textual data and extracts important linguistic features. It compares student answers with reference answers to calculate similarity scores. The authors used machine learning algorithms to improve grading accuracy. The research shows that automated systems can reduce human workload and improve grading consistency. The system also provides faster evaluation compared to manual grading. The study highlights the importance of semantic analysis in evaluating descriptive answers. The authors discuss the challenges of handling different writing styles and vocabulary variations. Experimental results show that the proposed system achieves reliable grading performance. The research contributes to the development of intelligent educational tools. It also demonstrates the effectiveness of NLP in automated answer evaluation.

The proposed automated grading system can be enhanced by integrating advanced NLP techniques such as deep learning and contextual embeddings. The study emphasizes the importance of feature extraction methods like tokenization, stemming, and part-of-speech tagging in improving system performance. It also highlights that semantic similarity plays a key role in evaluating descriptive answers accurately. The research suggests that future systems can incorporate adaptive learning mechanisms to improve grading over time. It discusses the importance of handling synonyms and paraphrased answers effectively. The paper also indicates that automated systems can be integrated with online learning platforms for real-time evaluation. It highlights the need for large and diverse datasets to train robust models. It also suggests providing feedback along with scores to enhance student learning. Overall, this research contributes to the development of efficient and scalable automated grading systems.

2.3 TEXT SIMILARITY DETECTION USING NLP TECHNIQUES

Daniel Cer, et al. (2018) research focuses on detecting similarity between texts using advanced NLP techniques. The authors proposed models that convert sentences into semantic vector representations. These vectors capture the contextual meaning of sentences. The system then calculates similarity scores between sentences using mathematical methods. The research demonstrates improved performance in text comparison tasks. These techniques can be applied in automated grading systems to evaluate student answers. The paper highlights the importance of semantic understanding in text processing. It also explains how NLP models can improve the accuracy of similarity detection. The study shows that deep learning models provide better results compared to traditional methods. The research contributed to modern NLP applications in education and assessment systems.

It explains that semantic vector representations help in capturing deeper meaning beyond surface-level text. The study highlights the importance of contextual embeddings in improving similarity accuracy. It suggests that advanced deep learning models such as sentence transformers can further enhance performance. The paper also discusses applications such as plagiarism detection and document clustering. It indicates that similarity detection techniques can be combined with other NLP methods for better results. The research highlights challenges such as handling polysemy and synonymy in language. It also explains the role of evaluation metrics in measuring system performance. The study suggests future improvements using hybrid approaches combining statistical and neural methods. Overall, this work contributes significantly to modern NLP-based evaluation systems.

2.4 ATTENTION IS ALL YOU NEED

Vaswani, A., Shazeer, N., Parmar, N., et al. (2017) proposed the Transformer architecture for natural language processing tasks. The authors introduced a model that relies entirely on attention mechanisms to process sequential data without using traditional recurrent or convolutional networks. The Transformer model allows the system to understand relationships between words in a sentence more effectively by analyzing the context of each word. This architecture significantly improved performance in various NLP applications such as machine translation, text classification, and semantic similarity analysis. The study demonstrates that attention-based models can process textual information more efficiently and accurately than earlier models. These techniques are widely used in modern NLP systems to understand sentence meaning and contextual relationships between words. The research contributed greatly to the development of advanced language models used in text analysis and automated evaluation systems.

It highlights that attention mechanisms allow models to focus on relevant parts of a sentence, improving contextual understanding. The research explains that Transformers overcome the limitations of sequential processing in RNNs. It also discusses the efficiency of parallel processing in reducing training time. The study suggests that transformer-based models can be scaled for large datasets and complex tasks. It highlights applications such as language translation, question answering, and text summarization. The paper also explains the role of self-attention in capturing long-range dependencies. It discusses the importance of pre-trained models like BERT and GPT in modern NLP systems. The research indicates challenges such as high computational cost. Overall, this work forms the backbone of current advanced NLP technologies.

2.5 SEMANTIC SIMILARITY MEASUREMENT USING WORD EMBEDDINGS

Tomas Mikolov, Kai Chen, et al., (2017) focuses on measuring semantic similarity between words and sentences using word embedding techniques. The authors introduced models that convert text into numerical vectors representing semantic meaning. These vectors allow machines to understand relationships between words and sentences. The paper explains how semantic similarity can be calculated using vector similarity measures such as cosine similarity. The approach helps in comparing texts based on meaning rather than exact word matches. This technique is widely used in NLP applications such as question answering and answer evaluation systems. The study shows that word embeddings capture contextual relationships between words effectively. It also demonstrates improved performance in text comparison tasks. The research contributed to the development of modern semantic analysis techniques. These techniques are useful in automated grading systems for evaluating descriptive answers. The paper highlights the importance of semantic representation in text processing. Overall, the study improved the accuracy of NLP-based evaluation systems.

Furthermore, the study highlights that word embedding techniques are essential for representing text in numerical form. It explains that embeddings capture semantic relationships between words effectively.It emphasizes that these techniques improve the performance of various NLP tasks. The study also highlights the importance of dimensionality reduction in handling large datasets. It suggests that combining embeddings with deep learning models can enhance accuracy. The paper explains the role of cosine similarity in comparing vector representations. It also discusses challenges such as handling out-of-vocabulary words. The research indicates that contextual embeddings can further improve results. Overall, this work plays a key role in semantic analysis and automated evaluation systems.

2.6 MACHINE LEARNING APPROACHES FOR AUTOMATED ESSAY EVALUATION

Mark D. Shermis and Jill Burstein, (2017) discusses machine learning approaches for evaluating essays automatically. They explain how machine learning models can analyze writing style, grammar, and content quality. These models are trained using datasets of graded essays to learn evaluation patterns. The system evaluates essays based on multiple linguistic features. The research highlights the benefits of automated grading systems in education. These systems improve grading efficiency and consistency. The authors also discuss challenges such as evaluating creativity and logical reasoning. Experimental results show that automated systems can produce reliable grading results. The research demonstrates the potential of machine learning in educational assessment. The study contributed to the development of advanced automated evaluation systems.

It explains that models can be trained using large datasets of previously graded essays to learn scoring patterns. The research emphasizes the importance of features such as grammar, coherence, vocabulary usage, and content relevance. It also discusses how supervised learning techniques improve the accuracy of evaluation. The study suggests that combining machine learning with NLP techniques can enhance overall system performance. It highlights the importance of preprocessing steps such as tokenization and normalization. The paper also addresses challenges in evaluating creativity and logical flow in essays. It indicates that hybrid models can be used to overcome these limitations. The research suggests future improvements using deep learning approaches. Overall, this work contributes to the development of reliable and efficient essay evaluation systems.

2.7 AUTOMATED ESSAY SCORING USING NATURAL LANGUAGE PROCESSING

Shermis M. D. and Burstein J. (2016) discussed the development of automated essay scoring systems using Natural Language Processing techniques. They explained how NLP can analyze grammar, sentence structure, and vocabulary to evaluate written responses. The study focuses on machine learning models trained using large datasets of previously graded essays, allowing the system to learn patterns from human grading and apply them to new essays. The system compares student responses with reference answers and evaluates them using multiple linguistic features. The paper highlights the advantages of automated grading systems in saving time and improving evaluation consistency, while also addressing challenges such as handling creative writing and variations in writing style. The research shows that automated systems can achieve accuracy comparable to human graders and emphasizes the importance of semantic understanding rather than simple keyword matching in modern evaluation systems.

It explains that these systems analyze various linguistic features such as syntax, semantics, and discourse structure. The research highlights that training models on large annotated datasets helps achieve better performance. It also discusses the importance of semantic understanding in evaluating descriptive answers. The paper suggests that integrating deep learning techniques can further enhance scoring accuracy. It highlights the role of feedback mechanisms in improving student learning outcomes. The study also addresses challenges such as bias and fairness in automated grading. It indicates that continuous model updates are required for better performance. The research suggests the use of hybrid approaches combining rule-based and machine learning methods. Overall, this work supports the advancement of intelligent grading systems.

2.8 SENTENCE SIMILARITY USING COSINE SIMILARITY

Li Y, Bandar Z, et al., (2016) research focuses on measuring sentence similarity using cosine similarity techniques. The authors proposed a method to represent sentences as numerical vectors. Cosine similarity is then used to measure the similarity between these vectors. The similarity score indicates how closely two sentences match in meaning. This technique is widely used in NLP applications including text comparison and information retrieval. The study demonstrates that cosine similarity provides accurate results in semantic comparison tasks. The method helps systems evaluate answers based on meaning rather than exact wording. The research also explains how sentence similarity can improve automated answer evaluation systems. The approach reduces errors caused by variations in sentence structure. The study shows that semantic similarity methods are effective for grading descriptive answers. This technique is commonly used in modern NLP evaluation systems.

It explains that representing sentences as vectors allows for mathematical comparison of their meanings. The study emphasizes that cosine similarity helps reduce errors caused by variations in wording. It also discusses how this method is widely used in information retrieval and text comparison tasks. The paper suggests that combining cosine similarity with advanced NLP models can improve accuracy. It highlights the importance of preprocessing techniques for better results. The research also explains challenges such as handling complex sentence structures. It indicates that integrating contextual embeddings can further enhance similarity detection. The study suggests future work in improving semantic representation techniques. Overall, this work contributes to improving automated answer evaluation systems.

2.9 SHORT ANSWER GRADING USING MACHINE LEARNING TECHNIQUES

Mohler M, and Mihalcea R (2015) research on grading short descriptive answers using machine learning techniques. The authors developed a system that compares student answers with reference answers using semantic similarity measures. The system uses NLP techniques to process and analyze textual data. It extracts important features from the text and calculates similarity scores between answers. The similarity score determines the marks assigned to each answer. The paper demonstrates that machine learning models can effectively automate the grading process. It also highlights the importance of semantic understanding in evaluating answers. The authors discuss challenges such as variations in sentence structure and vocabulary. The research shows that automated systems can achieve reliable evaluation results. The system improves grading speed and reduces manual workload for teachers. The study contributed to the development of automated short answer evaluation systems. It also improved the efficiency of digital assessment systems.

Furthermore, the study highlights that machine learning techniques are highly effective in grading short answers automatically. It explains that these systems rely on semantic similarity measures to compare student responses with reference answers. The research emphasizes the importance of feature extraction in analyzing textual data. It also discusses how models can be trained to handle variations in sentence structure and vocabulary. The study highlights the benefits of automated grading in reducing manual workload. It suggests that integrating NLP techniques improves the accuracy of evaluation. The paper also addresses challenges such as handling ambiguous and incomplete answers. It indicates that large datasets are required for training robust models. The research suggests future improvements using deep learning approaches. Overall, this work enhances the efficiency of digital assessment systems.

2.10 AUTOMATIC SHORT ANSWER GRADING SYSTEM

Leacock C, and Chodorow M., (2014) presents an automated system designed for grading short answers in educational assessments. The authors used NLP techniques to analyze student responses and compare them with model answers. The system focuses on identifying semantic similarity between answers rather than exact word matching. The research explains how linguistic features such as sentence structure and word meaning are used in evaluation. The system calculates similarity scores and assigns marks automatically. The authors highlight the importance of automated grading in large-scale assessments. The system reduces the workload of teachers and ensures faster result generation. The paper also discusses challenges such as handling synonyms and paraphrased answers. Experimental results show that automated grading systems can achieve high accuracy levels. The study contributed to the advancement of NLP-based evaluation systems. It demonstrated the practical use of NLP in educational technology.

It explains that NLP techniques help in analyzing student responses effectively. The research emphasizes the importance of semantic similarity in evaluating answers accurately. It also discusses how linguistic features such as syntax and word meaning contribute to grading. The paper suggests that integrating advanced machine learning models can improve system performance. It highlights the importance of handling synonyms and paraphrased answers correctly. The study also addresses challenges such as maintaining fairness and accuracy in grading. It indicates that providing detailed feedback along with scores can improve student learning. The research suggests future improvements using deep learning and contextual models. Overall, this work contributes to the development of efficient automated evaluation systems.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

In the traditional education system, the evaluation of descriptive answers is mainly performed manually by teachers or examiners. Teachers read each student's answer sheet carefully and assign marks based on their understanding and judgment. This process requires a significant amount of time and effort, especially when the number of students is large. Manual evaluation may also lead to inconsistency because different evaluators may interpret answers differently. Sometimes, students who write correct answers using different words or sentence structures may not receive proper marks.

The existing systems often rely on simple keyword matching techniques rather than understanding the actual meaning of the answer. As a result, answers that contain the correct keywords may receive marks even if the explanation is not accurate. Another limitation is that manual evaluation is prone to human errors and bias, which can affect the fairness of the grading process. Additionally, evaluating large volumes of answer sheets can delay the announcement of results. In modern education systems where online learning and digital examinations are increasing, traditional evaluation methods become inefficient. Therefore, the existing system lacks automation, semantic understanding, and efficiency in handling descriptive answer evaluation.

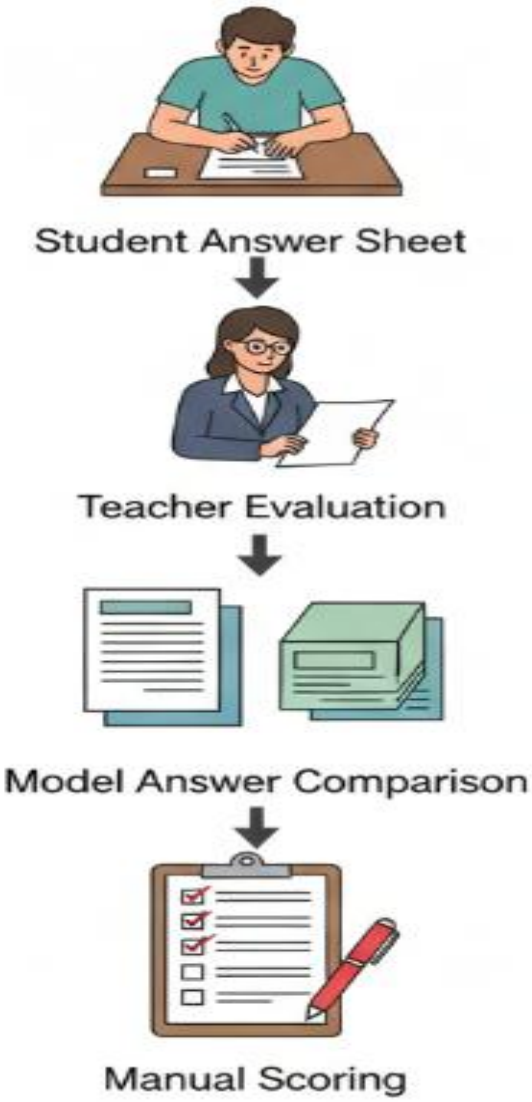

Fig 3.1 Existing System Architecture

3.2 DRAWBACKS:

- Time-consuming evaluation process
- Requires a lot of manual effort from teachers
- Possibility of human errors while grading
- Different evaluators may give different marks
- Delay in publishing examination results.

CHAPTER 4

PROBLEMS IDENTIFIED

The current method of evaluating descriptive answers in examinations mainly depends on manual checking by teachers. This approach requires a significant amount of time and effort, especially when a large number of students appear for exams. Teachers must carefully read each answer sheet and assign marks based on their understanding. Due to this manual process, inconsistencies in grading may occur, as different evaluators may interpret the same answer in different ways. In some cases, correct answers written using different words or sentence structures may not receive proper marks if the evaluator focuses only on specific keywords.

Manual evaluation also increases the chances of human errors. Teachers may overlook important points in a student's answer or assign incorrect marks due to fatigue or lack of concentration. Calculation mistakes can occur while totaling marks, which further affects the accuracy of the evaluation. Since the entire process depends on human judgment, such errors are difficult to avoid and may lead to unfair results and dissatisfaction among students.

Another major issue is the lack of consistency in grading. Different teachers may assign different marks for the same answer based on their personal interpretation and experience. Even the same teacher may evaluate similar answers differently at different times. This lack of uniformity reduces the reliability of the evaluation system and creates confusion and unfairness among students.

Evaluating a large number of answer sheets manually is also very difficult and time-consuming. Educational institutions often conduct examinations for hundreds or thousands of students, making the evaluation

process slow and difficult to manage. As a result, there is often a delay in publishing results, which affects academic planning and student progress.

Traditional evaluation methods often rely on keyword-based assessment rather than understanding the actual meaning of the answer. This leads to situations where answers containing expected keywords receive marks even if the explanation is incomplete, while correct answers written in different words may not be properly evaluated. This approach limits the effectiveness and fairness of the evaluation process.

Manual evaluation may also be influenced by human bias. Personal opinions, assumptions, or preferences of the evaluator can affect the grading process. In some cases, students may receive higher or lower marks due to subjective judgment rather than actual performance. This reduces transparency and fairness in the evaluation system.

With the increasing adoption of online education and digital learning platforms, traditional evaluation methods are becoming less practical. Online examinations require fast, scalable, and automated evaluation systems to handle large volumes of student submissions efficiently. Manual checking cannot meet these requirements, making it unsuitable for modern educational environments. Therefore, there is a clear need for an automated system that can evaluate descriptive answers quickly, accurately, and fairly by understanding the semantic meaning of the text.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM

The proposed system provides an automated solution for evaluating descriptive answers using Natural Language Processing techniques. Unlike traditional manual evaluation methods, the system reduces human effort and ensures faster and more consistent grading.

In this system, the faculty member first enters the student details such as student name and register number through the user interface. The faculty can then select the subject for which the evaluation is to be performed. After selecting the subject, the faculty uploads the model answer (answer key) that will be used as the reference for evaluation. The student answer file is also uploaded through the system interface.

Once the inputs are provided, the system processes the text using Natural Language Processing techniques. The uploaded answers undergo text pre-processing and feature extraction, and the similarity between the student answer and the model answer is calculated using cosine similarity. Based on the similarity score obtained, the system automatically generates the evaluation result.

Finally, the calculated score is displayed to the user and stored in an Excel sheet for maintaining evaluation records. This enables faculty members to easily manage student results and access them later for analysis or reporting.

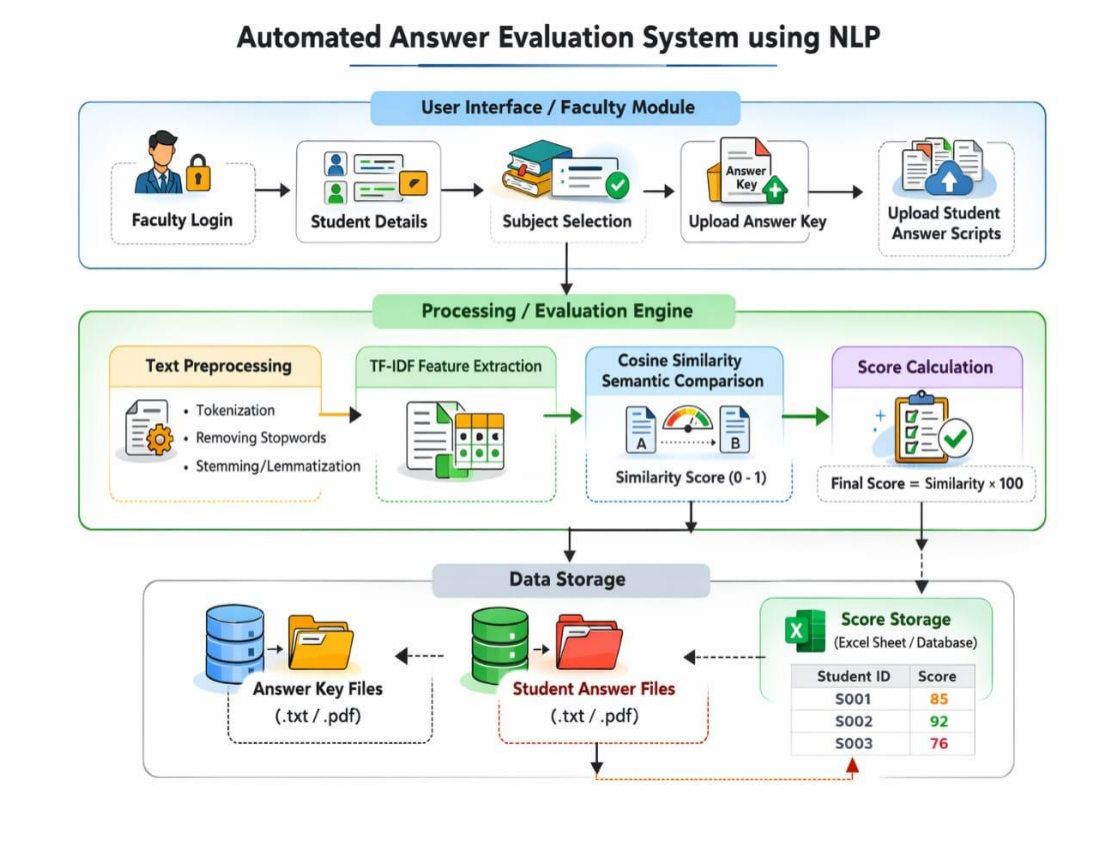

Fig 5.1 Architecture Diagram of Evaluation System

5.2 WORK FLOW

The overall working process of the automated answer evaluation system begins with the student answer being provided as input to the system. The text pre-processing module cleans and prepares the text for analysis. Next, the feature extraction module converts the text into meaningful numerical representations using NLP techniques. The semantic similarity module then compares the student answer with the model answer using cosine similarity. Based on the similarity score obtained, the system calculates the marks and generates the final evaluation result. This process ensures fast, accurate, and unbiased evaluation of descriptive answers.

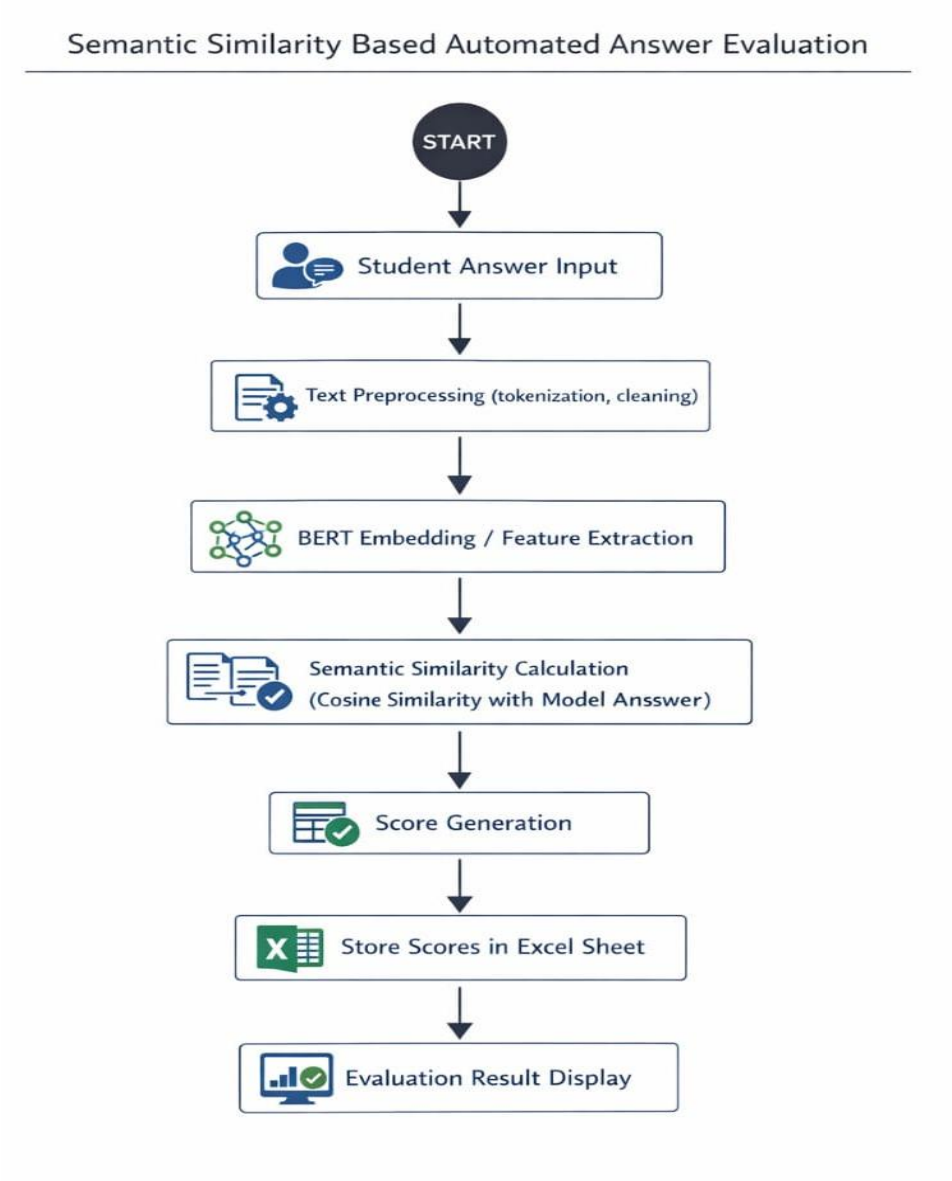

Fig 5.2 Workflow of the Proposed System

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

1. **Processor:** Intel i5 / Ryzen 5 or above
2. **RAM:** Minimum 8 GB
3. **Hard Disk:** 256 GB
4. **System Type:** 64-bit Architecture

6.2 SOFTWARE REQUIREMENTS

1. **Operating System:** Windows / Linux
2. **Programming Language:** Python
3. **Frontend:** HTML, CSS, JavaScript
4. **Backend:** Flask
5. **NLP Libraries:** NLTK, spaCy, Scikit-learn, Sentence-Transformers
6. **Database:** MySQL / SQLite
7. **IDE:** VS Code / PyCharm.

CHAPTER 7

SYSTEM IMPLEMENTATIONS

The system implementation describes how the proposed automated answer evaluation system is developed and executed using different technologies and modules. The system is implemented using Natural Language Processing (NLP) and machine learning techniques to analyze and evaluate descriptive answers automatically. In this system, the user first provides the student answer through a web interface. The system then processes the input text using various NLP techniques such as tokenization, stop-word removal, and lemmatization to clean and prepare the text for further analysis. After preprocessing, the system extracts important semantic features from the text using TF-IDF vectorization to convert text into numerical feature vectors.

These features represent the meaning of the sentences in numerical form. The system then calculates the similarity between the student answer and the model answer using cosine similarity. Based on the similarity score obtained, the system assigns marks automatically and generates the final evaluation result. The implementation of the system ensures accurate, fast, and unbiased evaluation of descriptive answers. It also reduces the manual workload of teachers and improves the efficiency of the grading process in educational institutions.

7.1 LIST OF MODULES

The system is implemented as a web-based application where faculty members first log in to access the evaluation platform. After logging in, the user enters student details such as student name and register number and selects the

subject for evaluation. The faculty uploads the model answer (answer key) and the student answer file through the interface. The system processes the uploaded files, evaluates the similarity between answers, generates marks automatically, and stores the results in an Excel sheet for record management.

To perform these functions efficiently, the system is divided into several modules, each responsible for a specific task in the evaluation process.

- User Interface Module
- Text Preprocessing Module
- Feature Extraction Module
- Semantic Similarity Evaluation Module
- Scoring and Result Generation Module

7.1.1 USER INTERFACE MODULE

The User Interface Module provides an interactive platform for teachers or evaluators to use the system easily. The system allows faculty members to enter student details such as student name and register number along with the student answers and model answers through a simple web interface. It also displays the evaluation results clearly, including the marks assigned and similarity score. The interface helps teachers manage the evaluation process efficiently without requiring technical knowledge. A well-designed interface improves the usability and effectiveness of the system.

7.1.2 TEXT PREPROCESSING MODULE

The Text Preprocessing Module prepares the input text for further analysis by cleaning and organizing the data. It removes unnecessary elements such as punctuation marks, special characters, and extra spaces from the text. The module also converts all words into lowercase to maintain consistency. Tokenization is used to split sentences into individual words or tokens. Lemmatization is applied to convert words into their base forms to improve text analysis. This preprocessing stage helps improve the accuracy of the similarity calculation by ensuring that the text data is properly formatted.

7.1.3 FEATURE EXTRACTION MODULE

The Feature Extraction Module converts the processed text into numerical representations that can be understood by machine learning models. In this system, advanced language models such as BERT embeddings are used to capture the semantic meaning of sentences. These embeddings represent words and sentences as vectors in a multidimensional space. By using these numerical representations, the system can understand the contextual meaning of the text. This module plays an important role in enabling the system to compare answers based on their meaning rather than exact word matches.

7.1.4 SEMANTIC SIMILARITY EVALUATION MODULE

The Semantic Similarity Evaluation Module compares the student answer with the model answer to determine how closely they match in meaning. The system uses cosine similarity to measure the similarity between the vector representations of the two texts. The similarity score ranges from 0 to 1, where a higher value indicates a stronger similarity between the answers. This module ensures that the evaluation focuses on semantic meaning rather than exact

wording. It allows the system to recognize correct answers even when students use different sentence structures or vocabulary.

7.1.5 SCORING AND RESULT GENERATION MODULE

The Scoring and Result Generation Module assigns marks to the student answer based on the similarity score calculated by the system. The module converts the similarity value into marks according to predefined evaluation criteria and categorizes the answer as correct, partially correct, or incorrect based on the similarity level. After calculating the marks, the system displays the evaluation results to the faculty through the user interface. In addition, the generated scores and student details are automatically stored in an Excel sheet for record management and future reference. This module ensures that the evaluation process is fast, accurate, and consistent while helping teachers review and manage grading results efficiently.

CHAPTER 8

SYSTEM TESTING

System testing is an important phase in software development used to verify whether the developed system functions correctly and meets the required objectives. The Semantic and Similarity-Based Answer Evaluation System was tested to ensure reliability, accuracy, and efficiency. During testing, the system functionalities such as faculty login, student detail entry, subject selection, answer key upload, student answer upload, automatic evaluation, and Excel-based score storage were verified. The testing process ensured that all system modules interacted properly and produced accurate evaluation results. Various testing techniques such as unit testing, integration testing, system testing, performance testing, security testing, and usability testing were conducted to identify possible errors and confirm that the system performs as expected. Through these testing procedures, the system was validated to evaluate student answers correctly and operate efficiently before final deployment.

8.1 UNIT TESTING

Unit testing is performed to test individual modules or components of the system separately. Each module of the system such as the user interface module, text pre-processing module, feature extraction module, semantic similarity evaluation module, and scoring module is tested independently. The main objective of unit testing is to verify that each module works correctly according to the design. During this process, inputs are provided to each module and the outputs are checked to ensure that the expected results are obtained. Unit testing helps identify errors at an early stage of development and makes debugging easier.

8.2 INTEGRATION TESTING

Integration testing is performed after unit testing to verify the interaction between different modules of the system. In this phase, multiple modules are combined and tested together to ensure that they work properly as a complete system. For example, the text preprocessing module is integrated with the feature extraction module and similarity evaluation module. The system is tested to check whether the output of one module correctly becomes the input for the next module. This testing process helps identify issues related to module communication and data transfer.

8.3 SYSTEM TESTING

System testing is performed to evaluate the complete and integrated system as a whole. In this stage, the entire application is tested to ensure that all functionalities work correctly according to the project requirements. The system is tested with different inputs to verify whether the evaluation process works properly. For example, student answers are provided as input and the system generates similarity scores and marks automatically. System testing ensures that the system performs the evaluation accurately and produces consistent results. This testing phase confirms that the system meets the functional and technical requirements of the project.

8.4 PERFORMANCE TESTING

Performance testing is used to evaluate how well the system performs under different conditions. It checks the speed, responsiveness, and efficiency of the system while processing student answers. The system is tested with multiple inputs to determine whether it can handle large numbers of answers without slowing down. Performance testing ensures that the system processes

answers quickly and provides results within a short time. This testing also helps identify any performance issues such as slow processing or high memory usage. By improving these factors, the system can operate efficiently even when handling large datasets.

8.5 SECURITY TESTING

Security testing ensures that the system is protected from unauthorized access and data misuse. Since the system handles student answers and evaluation results, it is important to ensure data confidentiality and integrity. Security testing checks whether the system properly restricts access to authorized users only. It also verifies that user data is stored securely and protected from potential threats. By implementing proper security measures, the system ensures safe and reliable operation.

8.6 USABILITY TESTING

Usability testing is conducted to evaluate how easy and user-friendly the system is for teachers or evaluators. The system interface is tested to ensure that users can easily upload student answers and view evaluation results. The design of the user interface should be simple, clear, and easy to understand. Usability testing helps identify any difficulties users may face while interacting with the system. Feedback from users can be used to improve the interface and overall user experience. A user-friendly system increases efficiency and makes the evaluation process more convenient for teachers.

CHAPTER 9

RESULT AND DISCUSSION

9.1 RESULT

The results of the proposed Semantic and Similarity-Based Answer Evaluation System demonstrate its effectiveness in automatically evaluating descriptive answers. The system successfully processed student answers using NLP techniques such as tokenization, stop word removal, and lemmatization. After preprocessing, the answers were converted into numerical representations for similarity analysis.

The system computed similarity scores between student answers and model answers using cosine similarity and assigned marks accordingly. It was able to accurately classify answers as correct, partially correct, or incorrect. The evaluation process was completed within a few seconds, showing a significant improvement in speed compared to manual evaluation.

Additionally, the system generated organized output by storing student details and marks in an Excel sheet, making it easier for teachers to manage and review results. The results indicate that the system provides fast, accurate, and consistent evaluation of descriptive answers.

9.2 DISCUSSION

The discussion highlights the overall performance and impact of the proposed system in real-world educational scenarios. The system reduces the burden on teachers by automating the evaluation process and significantly saving time. Unlike traditional methods, the system ensures consistency in grading by applying the same evaluation criteria to all answers.

The use of semantic similarity allows the system to understand the meaning of answers rather than relying only on exact word matches. This makes the evaluation more flexible and fair, as students can express answers in different ways. The system also minimizes human bias, ensuring objective and transparent assessment.

Furthermore, the ability to handle large volumes of answer scripts makes the system suitable for online examinations and digital learning platforms. Overall, the discussion shows that the proposed system enhances efficiency, accuracy, and fairness in the evaluation process, making it a valuable tool for modern education systems.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The Semantic and Similarity-Based Answer Evaluation System using Natural Language Processing provides an effective solution for evaluating descriptive answers automatically. In traditional examination systems, teachers manually evaluate answer sheets, which requires significant time and effort. Manual evaluation may also lead to inconsistencies and human bias in grading. The proposed system addresses these problems by introducing an automated evaluation method based on semantic similarity.

The system uses NLP techniques to preprocess the input text and convert it into a meaningful format for analysis. Feature extraction methods such as BERT embeddings help capture the semantic meaning of the sentences. The similarity between the student answer and the model answer is calculated using cosine similarity. Based on this similarity score, the system automatically assigns marks and categorizes the answers.

The proposed system improves the efficiency of the evaluation process by reducing the time required for grading. It also ensures fairness and consistency in marking because the evaluation is performed by predefined algorithms rather than human judgment. The system can handle a large number of student answers, making it suitable for modern online examinations and digital learning platforms.

Overall, the developed system demonstrates that NLP and machine learning techniques can be effectively used in educational assessment. The system reduces manual workload, improves grading accuracy, and supports automated evaluation in educational institutions. In the future, the system can

be enhanced by adding features such as grammar analysis, detailed feedback generation, and integration with online learning platforms. These improvements can make the system more advanced and useful for modern education systems.

10.2 FUTURE ENHANCEMENTS

Integration with Online Examination Systems

The system can be integrated with online examination platforms so that student answers submitted during digital exams can be automatically evaluated. This will help educational institutions conduct large-scale online tests and receive instant evaluation results.

Grammar and Spelling Checking

Future versions of the system can include grammar and spelling analysis features. This will allow the system to evaluate not only the correctness of the content but also the quality of writing in student answers

Automatic Feedback Generation

The system can be enhanced to provide detailed feedback to students based on their answers. This feedback will help students understand their mistakes and improve their knowledge and writing skills.

Support for Multiple Languages

Currently, many systems evaluate answers only in English. In the future, the system can be improved to support multiple languages so that students can write answers in different languages while still receiving accurate evaluation.

Mobile Application Development

A mobile-based application can be developed so that teachers can easily upload answers and view evaluation results using smartphones or tablets. This will make the system more accessible and convenient.

Integration with Learning Management Systems (LMS)

The system can be integrated with popular Learning Management Systems such as Moodle or Google Classroom. This will allow teachers to manage assignments, exams, and evaluation results within a single platform.

APPENDIX – A

SOURCE CODE

```
from flask import Flask, render_template, request, redirect, url_for, session

import pandas as pd

import pytesseract

import cv2

import os

import re

from pdf2image import convert_from_path

from docx import Document

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics.pairwise import cosine_similarity

app = Flask(__name__)

app.secret_key = "secretkey"

UPLOAD_FOLDER = "uploads"

EXCEL_FILE = "subjects.xlsx"

os.makedirs(UPLOAD_FOLDER, exist_ok=True)


# ---------- GET SUBJECT LIST ----------

def get_subjects():
```

```
    if not os.path.exists(EXCEL_FILE):
        return []
    xl = pd.ExcelFile(EXCEL_FILE)
    return xl.sheet_names

# ---------- CLEAN TEXT ----------
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z0-9\s]', '', text)
    return text

# ---------- TEXT EXTRACTION ----------
def extract_text(file_path):
    ext = os.path.splitext(file_path)[1].lower()
    text = ""
    if ext == ".txt":
        with open(file_path, "r", encoding="utf-8") as f:
            text = f.read()
    elif ext == ".docx":
        doc = Document(file_path)
        for para in doc.paragraphs:
```

```
            text += para.text
    elif ext == ".pdf":
        images = convert_from_path(file_path)
        for img in images:
            text += pytesseract.image_to_string(img)
    elif ext in [".png", ".jpg", ".jpeg"]:
        img = cv2.imread(file_path)=
        text = pytesseract.image_to_string(img)
    elif ext == ".xlsx":
        df = pd.read_excel(file_path)
        for col in df.columns:
            text += " ".join(df[col].astype(str).tolist())
    return text


# ---------- EVALUATION ----------
def evaluate_answer(student_text, answer_key):
    student_text = clean_text(student_text)
    answer_key = clean_text(answer_key)
    vectorizer = TfidfVectorizer()
    vectors = vectorizer.fit_transform([student_text, answer_key])
    similarity = cosine_similarity(vectors)[0][1]
```

```
    return round(similarity * 100, 2)


# ---------- SAVE RESULT ----------
def save_result(student, regno, subject, score):
    new_row = pd.DataFrame({
        "Student Name":[student],
        "Register Number":[regno],
        "Mark":[score]
    })
    if os.path.exists(EXCEL_FILE):
        with pd.ExcelWriter(EXCEL_FILE, engine="openpyxl", mode="a",
if_sheet_exists="overlay") as writer:
            try:
                existing = pd.read_excel(EXCEL_FILE, sheet_name=subject)
                updated = pd.concat([existing, new_row], ignore_index=True)


                updated.to_excel(writer, sheet_name=subject, index=False)
            except:
                new_row.to_excel(writer, sheet_name=subject, index=False)
# ---------- ROUTES ----------
@app.route("/")
```

```
def welcome():
    return render_template("welcome.html")
@app.route("/login", methods=["GET","POST"])
def login():
    if request.method == "POST":
        username = request.form["username"]
        password = request.form["password"]
        if username == "admin" and password == "admin123":
            session["user"] = username
            return redirect(url_for("index"))
        else:
            return render_template("login.html", error="Invalid login")
    return render_template("login.html")
@app.route("/evaluate", methods=["GET","POST"])
def index():
    if "user" not in session:
        return redirect(url_for("login"))
    subjects = get_subjects()
    score = None
    if request.method == "POST":
        student_name = request.form["student_name"]
```

```
        regno = request.form["regno"]
        subject = request.form["subject"]
        answer_key = request.files["answer_key"]
        student_ans = request.files["student_answer"]
        key_path = os.path.join(UPLOAD_FOLDER, answer_key.filename)
        student_path = os.path.join(UPLOAD_FOLDER, student_ans.filename)
        answer_key.save(key_path)
        student_ans.save(student_path)
        key_text = extract_text(key_path)
        student_text = extract_text(student_path)
        score = evaluate_answer(student_text, key_text)
        save_result(student_name, regno, subject, score)
    return render_template("index.html", subjects=subjects, score=score)
if __name__ == "__main__":
    app.run(debug=True)
```

APPENDIX – B

SCREENSHOTS

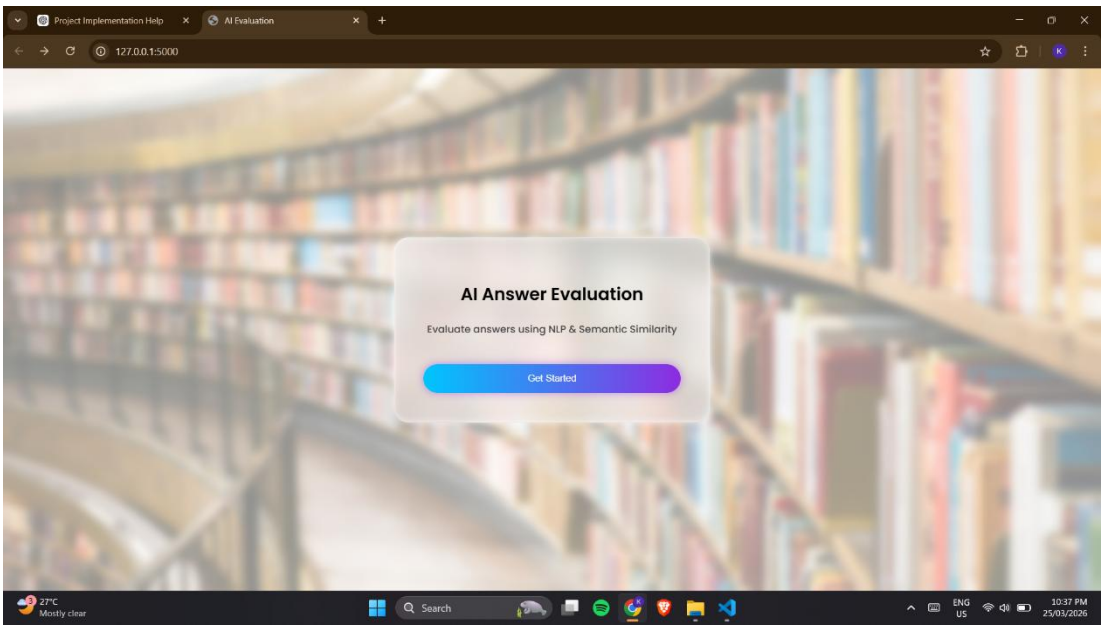

Figure. B.1 Get Access page

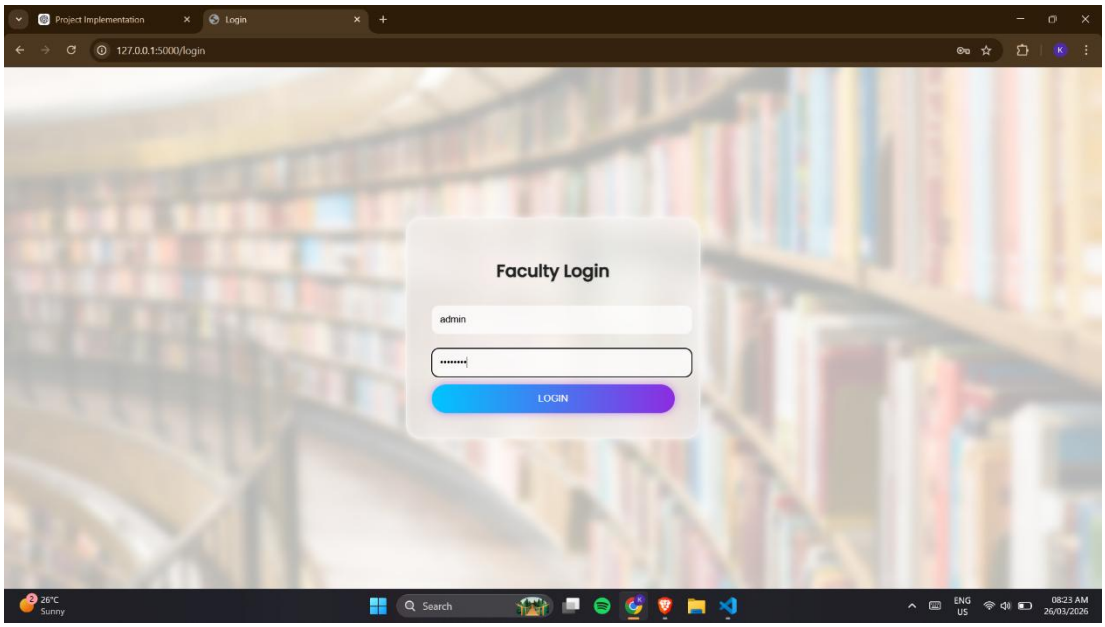

Figure. B.2 Login page

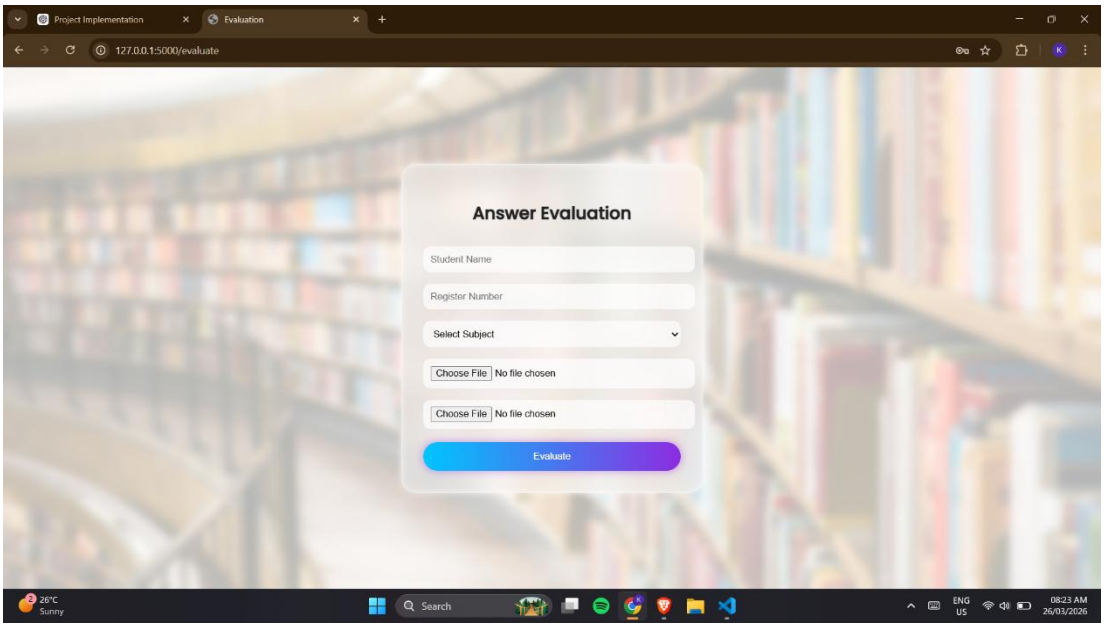

Figure. B.3 Answer Evaluation page

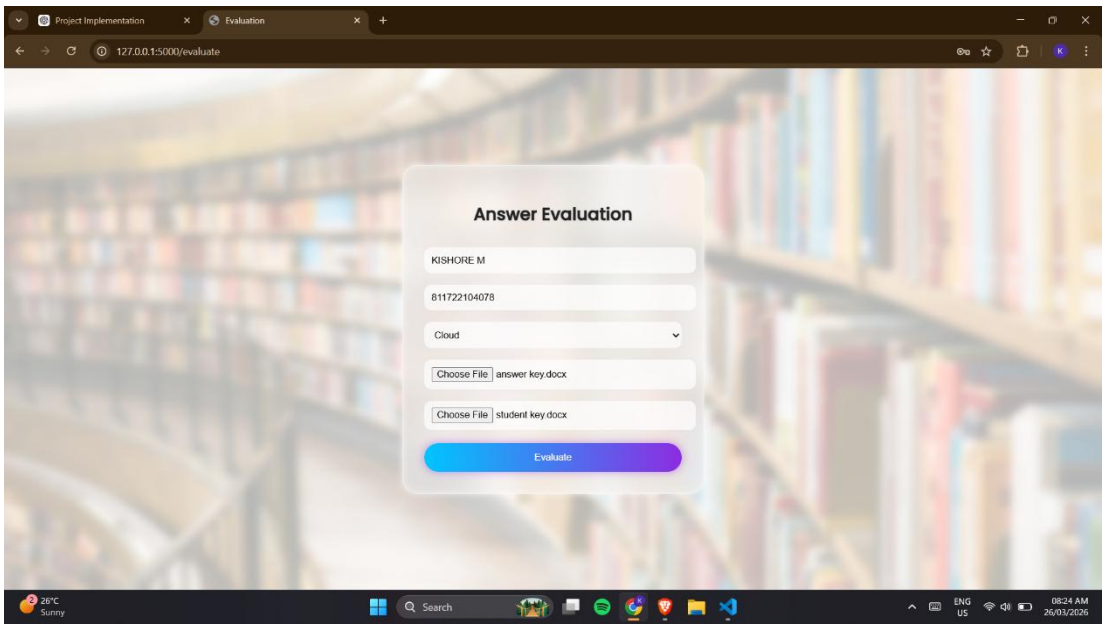

Figure. B.4 Data Insertion page

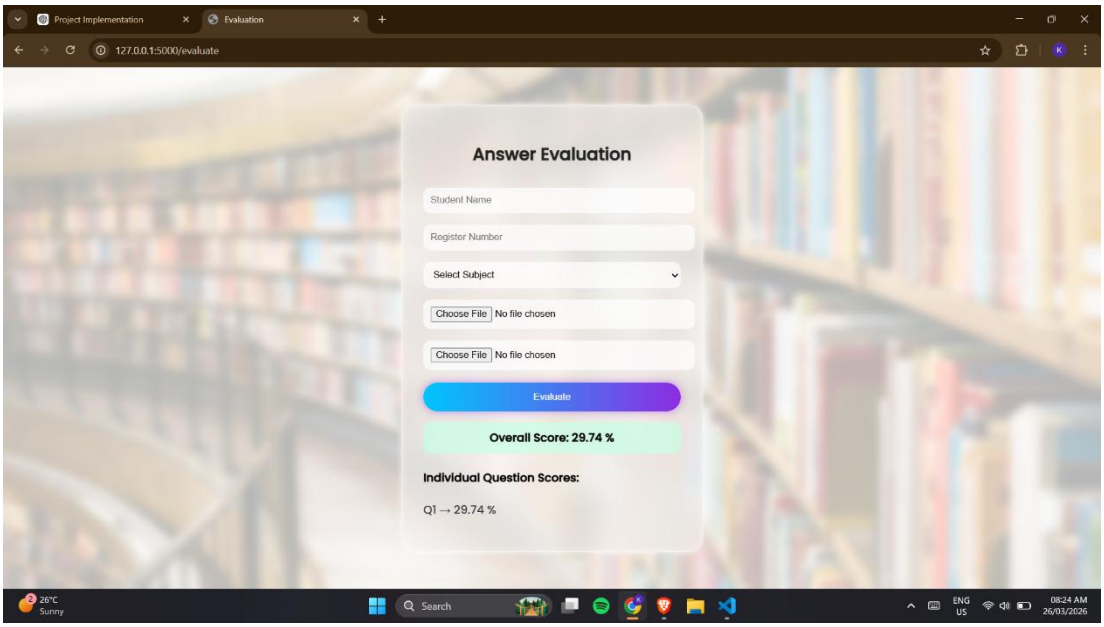

Figure. B.5 Processing / Output page

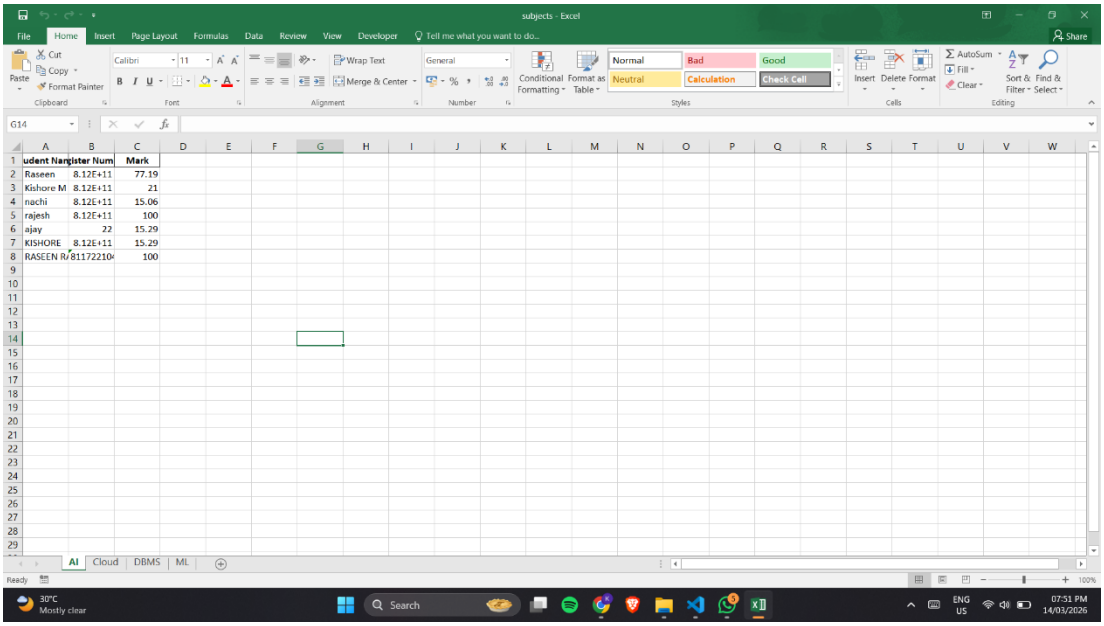

Figure. B.6 Database Collection

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